

Amartya Basu

CONTACT INFORMATION	NCS 330, Department of Computer Science Stony Brook University Stony Brook, NY, USA	Homepage: amartya-pixel.github.io E-mail: ambasu@cs.stonybrook.edu
EDUCATION	Stony Brook University , NY, USA Ph.D. in Computer Science Advisor: Dr. Shubham Jain Indian Institute of Technology Madras , Chennai, India M.S in Computer Science & Engineering Thesis: Mapping Pervasive Environments using Radio Tomography and Neural Radiance Field Government College of Engineering and Leather Technology , Kolkata, India B.Tech. in Computer Science & Engineering	2025 – present 2022 – 2025 2016 – 2020
AWARDS & ACHIEVEMENTS	- Travel Grant Award – ACM HotMobile 2026. - Research Excellence Award, IIT Madras. - Pre-Incubation Funding from NIRMAAN, IIT Madras (Jul–Nov 2024 cohort). - Institute Research (IR) Award, IIT Madras. (Sole recipient in the M.S. category, Jul–Nov 2024) - Travel Grant Award – IITB CSE Research Symposium. - Travel Grant Award – ACM SIGMETRICS / PERFORMANCE. - Secured 98.58 percentile $\approx$1 lakh candidates in GATE (CS).	2026 2024 2024 2023 2022 2020
RESEARCH WORK EXPERIENCE	Stony Brook University , NY, USA Research Assistant, Pervasive Computing and Smart Sensing (PiCASSo) Lab, Department of Computer Science. Indian Institute of Technology Madras , Chennai, India Research Assistant, SeNSE Lab, Department of Computer Science & Engineering. Tata Consultancy Services , Kolkata, India Assistant System Engineer. ITC Infotech , Kolkata, India Web Development Intern.	2025 – present 2022 – 2024 2020 – 2021 Summer 2019
PUBLICATIONS & PREPRINTS	Beyond-Voice: Leveraging Articulatory Motion for Next-Gen AI Assistants. Tanmay Srivastava, Amartya Basu , Shubham Jain. 27th International Workshop on Mobile Computing Systems and Applications. ACM HotMobile 2026 . SpecNeRF: Neural Radiance Field Driven Wireless Coverage Mapping for 5G Networks. Amartya Basu , Ayon Chakraborty. AIoT Workshop, ACM MobiHoc 2024 . Ubiquitous Indoor Mapping using Mobile Radio Tomography. Amartya Basu , Ayon Chakraborty, Kush Jajal. IEEE Transactions on Mobile Computing , Vol. 23, No. 12, 2024.	
PATENTS	A System and Method for Mapping Indoor Spaces in Real-time . Ayon Chakraborty and Amartya Basu . Granted – 567451 (India) .	
PRESS COVERAGE	Ubiquitous Indoor Mapping using Mobile Radio Tomography : Times of India , Hindustan Times , The Hindu , India Today .	

PROFESSIONAL
SERVICES

Journal Reviewer

- IEEE Transactions on Visualization and Computer Graphics — 2026, 2025.
- IEEE Transactions on Mobile Computing — 2026, 2025, 2024.
- Ad Hoc Networks (Elsevier) — 2025, 2024.

Artifact Evaluation Committee

- ACM International Conference on emerging Networking Experiments and Technologies (CoNEXT 2025).
- ACM International Conference on Mobile Systems, Applications, and Services (MobiSys 2025).
- ACM Conference on Embedded Networked Sensor Systems (SenSys 2024).

Conference Activities

- Web Chair of SOCIETY workshop in conjunction with ICDCN 2025, IIT Hyderabad, India.
- Volunteer for the ICDCN 2024 conference, IIT Madras, India.

Institute Activities

- Organizer of CSE Bits monthly event in the Department of CSE, IIT Madras — 2023.
- Subject matter expert for GATE CS NPTEL — 2022. [\[video\]](#)

TEACHING
ASSISTANTSHIPS

Smart Sensing for Internet of Things (*Graduate level*) Jul–Nov 2024
Instructor: [Dr. Ayon Chakraborty](#).

Advanced Data Structures and Algorithm (*Graduate level*) Jan–May 2024
Instructor: [Dr. C. Pandu Rangan](#).

Foundation for Computer System Design (*Undergraduate level*) Jul–Nov 2022
Instructor: [Dr. Ayon Chakraborty](#).